

Analysis of Meteorological Conditions in Spain by Means of Clustering Techniques

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Purpose of this document

This is an expanded literature-survey summary prepared for the project's mini-hackathon/review. It summarizes the published research rather than reproducing the copyrighted full paper.

1. Background and Problem

Meteorological datasets contain several measurements that jointly describe weather conditions. Looking at individual variables separately can make it difficult to characterize overall weather states. This paper investigates how clustering techniques can be used to analyze meteorological conditions and identify meaningful groupings.

2. Aim of the Study

The main aim is to provide a comprehensive analysis of clustering techniques when applied to real meteorological data. Rather than relying on one algorithm, the study compares multiple partitional and hierarchical approaches and uses several clustering evaluation measures to examine the quality of the resulting groups.

3. Dataset

The study uses real meteorological data recorded by the Spanish Meteorological Agency (AEMET) at four weather stations. The data covers six years between 2004 and 2010 and includes major daily meteorological variables. This gives the study a real-world basis rather than relying on a synthetic dataset.

4. Algorithms Studied

The paper examines several approaches, including K-means, K-medoids, SOM K-means, Agglomerative Hierarchical Clustering and clustering based on Gaussian Mixture Models. Different distance criteria are also considered. This makes the study useful for understanding that clustering is not a single technique and that different algorithms may produce different structures.

5. Evaluation Measures

A particularly useful aspect of the paper is its use of clustering evaluation criteria. The study discusses Calinski–Harabasz, Davies–Bouldin, Gap and Silhouette criteria. These measures help evaluate cluster compactness, separation or overall quality instead of judging a clustering result only by looking at a graph.

6. Findings and Interpretation

The study applies the clustering methods to the daily meteorological observations in order to characterize actual weather conditions at the selected Spanish stations. The comparison demonstrates the usefulness of both partitional and hierarchical methods for meteorological analysis and shows why evaluation measures are important when selecting or comparing clustering approaches.

7. Significance

The paper is especially relevant because it treats meteorological analysis as an unsupervised learning problem and compares multiple algorithms on the same type of data that our project uses. It also provides a methodological basis for comparing K-means and Hierarchical Clustering instead of implementing only one algorithm.

8. Connection to Our Project

Our project follows a similar high-level idea: use multiple environmental variables, apply unsupervised clustering and compare clustering approaches. The major difference is the geographic dataset. The paper studies Spanish weather-station data, whereas our project uses the Indian Climate Dataset (2024–2025) from Kaggle. We can also use clustering evaluation measures such as Silhouette Score and Davies–Bouldin Index, which are directly relevant to the methodology described in this paper.

9. Viva Points

If asked why this paper is important, say that it shows how several clustering algorithms can be applied to meteorological data and compared using objective evaluation measures. If asked what we borrowed from it, say the idea of comparing clustering methods and evaluating the resulting clusters using quantitative metrics.

Reference: See the publisher record/DOI for the original article.